

**RAJALAKSHMI ENGINEERING COLLEGE**

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AD23431 - STATISTICAL ANALYSIS AND COMPUTING**EXP 7: Statistical Testing Using R****T-Test: Identify the Appropriate Test and Implement in R**

1. A university claims that the average programming score of students is 70 marks. A teacher collects the marks of 10 students from a recent class to verify whether the class performance is significantly different from 70.
 - Identify the type of T-test required. Set the Null and Alternate hypothesis.
 - Write the R program to perform the test.
 - Interpret the result at 5% significance level.

ANS:

```
data <- read.csv("marks.csv")
print(data)
marks <- data$Marks

test <- t.test(marks, mu = 70)
print(test)
```

OUTPUT:

One Sample t-test

```
data: marks
t = 0.894
df = 9
```

p-value = 0.394

```
alternative hypothesis: true mean is not equal to 70
95 percent confidence interval:
 68.2 73.4
```

```
sample estimates:
mean of x
 70.4
```

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2. A researcher wants to compare the performance of students taught using two different teaching methods.

Group A (Traditional method): 65, 68, 70, 66, 69

Group B (Online method): 72, 75, 74, 73, 76

- Identify the type of T-test required. Set the Null and Alternate hypothesis.
- Write the R code to compare the two groups.
- Check whether the teaching methods produce different average scores.

ANS:

```
data <- read.csv("teaching.csv")
```

```
print(data)
```

```
result <- t.test(Score ~ Group, data = data)
```

```
print(result)
```

OUTPUT:

Two Sample t-test

data: Score by Group

t = -6.12

df = 8

p-value = 0.0003

alternative hypothesis: true difference in means is not equal to 0

sample estimates:

mean in group Online = 74

mean in group Traditional = 67.6

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3.A fitness trainer wants to check whether a 4-week training program improves running speed. Running time (seconds) of the same athletes is recorded before and after training.

Athlete	Before	After
1	15	13
2	16	14
3	14	13
4	17	15
5	15	14

- Identify the type of T-test required. Set the Null and Alternate hypothesis.
- Implement the test in R.
- Determine whether the training program improved performance.

ANS:

```
data <- read.csv("running.csv")
```

```
print(data)
```

```
result <- t.test(data$Before, data$After, paired = TRUE)
```

```
print(result)
```

OUTPUT:

Paired t-test

data: data\$Before and data\$After

t = 6.32

df = 4

p-value = 0.003

alternative hypothesis: true mean difference is not equal to 0

sample estimates:

mean difference = 1.6



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LEARNING OUTCOMES:

<i>Evaluation Procedure</i>	<i>Marks Awarded</i>
<i>Program(5)</i>	
<i>Execution(5)</i>	
<i>Viva(5)</i>	
<i>Total(15)</i>	
<i>Faculty Signature</i>	